Computação Científica para a Indústria · IFSC

Mosquitos pelo som

Detecção de Aedes aegypti
por análise acústica

Luiz Eduardo do Nascimento Ciarallo

INSTITUTO FEDERAL
Santa Catarina

INSTITUTO FEDERAL
Santa Catarina
Câmpus
Florianópolis

O áudio pode apoiar a vigilância do Aedes

Hoje: vistoria e coleta.
Identificação com lupa.
Proposta: analisar o voo.
Classificar o áudio.
Sinalizar presença de Aedes.
Apoiar o monitoramento.





Pneus e recipientes com água · SES-RS / SESA-ES

Triagem acústica para apoiar a vigilância.

INSTITUTO FEDERAL
Santa Catarina

Um áudio deve levar a uma de três classes

Entrada: áudio de voo.
Saída por gravação:
Aedes aegypti
Outro mosquito
Ruído / sem mosquito
Aprendizado supervisionado.



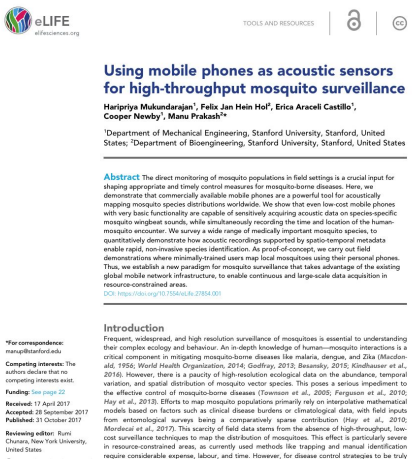
Aedes aegypti · fotografia CDC

Rótulos conhecidos ensinam as três classes.

INSTITUTO FEDERAL
Santa Catarina

A literatura orienta captura, memória e robustez

Mukundarajan · 2017



eLIFE
eLifesciences.org

TOOLS AND RESOURCES

Using mobile phones as acoustic sensors for high-throughput mosquito surveillance

Haripriya Mukundarajan[1], Felix Jan Hein Hol[2], Erica Araceli Castillo[1], Cooper Newby[1], Manu Prakash[2]*

[1]Department of Mechanical Engineering, Stanford University, Stanford, United States; [2]Department of Bioengineering, Stanford University, Stanford, United States

Abstract The direct monitoring of mosquito populations in field settings is a crucial input for shaping appropriate and timely control measures for mosquito-borne diseases. Here, we demonstrate that commercially available mobile phones are a powerful tool for acoustically mapping mosquito species distributions worldwide. We show that even low-cost mobile phones with very basic functionality are capable of sensitively acquiring acoustic data on species-specific mosquito wingbeat sounds, while simultaneously recording the time and location of the human-mosquito encounter. We survey a wide range of medically important mosquito species, to quantitatively demonstrate how acoustic recordings supported by spatio-temporal metadata enable rapid, non-invasive species identification. As proof-of-concept, we carry out field demonstrations where minimally-trained users map local mosquitoes using their personal phones. Thus, we establish a new paradigm for mosquito surveillance that takes advantage of the existing global mobile network infrastructure, to enable continuous and large-scale data acquisition in resource-constrained areas.

DOI: https://doi.org/10.7554/eLife.27854.001

Introduction

Frequent, widespread, and high resolution surveillance of mosquitoes is essential to understanding their complex ecology and behaviour. An in-depth knowledge of human—mosquito interactions is a critical component in mitigating mosquito-borne diseases like malaria, dengue, and Zika (*Macdonald, 1956*; *World Health Organization, 2014*; *Godfray, 2013*; *Besansky, 2015*; *Kindhauser et al., 2016*). However, there is a paucity of high-resolution ecological data on the abundance, temporal variation, and spatial distribution of mosquito vector species. This poses a serious impediment to the effective control of mosquito-borne diseases (*Townson et al., 2005*; *Ferguson et al., 2010*; *Hay et al., 2013*). Efforts to map mosquito populations primarily rely on interpolative mathematical models based on factors such as clinical disease burdens or climatological data, with field inputs from entomological surveys being a comparatively sparse contribution (*Hay et al., 2010*; *Mordecai et al., 2017*). This scarcity of field data stems from the absence of high-throughput, low-cost surveillance techniques to map the distribution of mosquitoes. This effect is particularly severe in resource-constrained areas, as currently used methods like trapping and manual identification require considerable expense, labour, and time. However, for disease control strategies to be truly

***For correspondence:** manup@stanford.edu

Competing interests: The authors declare that no competing interests exist.

Funding: See page 22

Received: 17 April 2017
Accepted: 28 September 2017
Published: 31 October 2017

Reviewing editor: Rumi Chunara, New York University, United States

Captura com celulares

Torres · 2025



Dengue-Infected Mosquito Detection with Uncertainty Evaluation of the Wingbeats using Monte Carlo Dropout

Israel Torres, Mariko Nakano, *Member, IEEE*, Jorge Cime-Castillo, Enrique Escamilla-Hernandez, *Member, IEEE*, Osvaldo Lopez-Garcia, and Humberto Lanz-Mendoza

***Abstract*— Considering that Aedes mosquitoes are a principal vector of the Dengue virus, which can cause the death of infected people in the worst case, accurate detection of infected Aedes mosquitoes is very important to prevent the further spread of the virus. This paper proposes a detection algorithm for infected *Aedes aegypti* mosquitoes by Dengue Virus-2 using their wingbeat signals. The proposed algorithm uses Long Short-Term Memory (LSTM) as a classifier of the input wingbeat signal into healthy mosquitoes and infected mosquitoes. All living beings, even those of the same species, have different characteristics depending on the season in which they are born, temperature, humidity, food, etc. Therefore, mosquitoes reared in the laboratory have individual differences although it is fed by the same blood infected. Considering these individual differences, an uncertainty measure based on Monte-Carlo dropout is introduced. The proposed algorithm detects approximately 5% of uncertainty data from all input wingbeat signals in the test set and provides a classification accuracy of 94.87% without any uncertainty.**

Link to graphical and video abstracts, and to code: https://latamt.ieeer9.org/index.php/transactions/article/view/9221

***Index Terms*—dengue fever, infected mosquitoes, Long Short-Term Memory, Monte Carlo dropout, uncertainty**

I. INTRODUCTION

CERTAIN mosquito species are responsible for the transmission of diseases such as dengue fever, Chikungunya fever, Zika, and malaria. Notably, *Aedes aegypti* and *Aedes albopictus* are the primary vectors of dengue fever, which can, in severe cases, result in mortality among infected individuals. According to a report from the Centers for Disease Control and Prevention (CDC) in the United States, approximately four billion people in the world are at risk of

The associate editor coordinating the review of this manuscript and approving it for publication was Sacha Friany (*Corresponding author: Mariko Nakano*).

I. Torres, M. Nakano, E. Escamilla-Hernandez, and O. Lopez-Garcia are with Instituto Politecnico Nacional, Culhuacan, Coyoacan, Mexico (e-mails: itorres@ipn.mx, mnakano@ipn.mx, and eescamillah@ipn.mx, olopezg@ipn.mx).

J. Cime-Castillo, and H. Lanz-Mendoza are with Instituto Nacional de Salud Publica, Cuernavaca, Morelos, Mexico (e-mails: jorge.cime@insp.mx, and hlanzm@insp.mx).

constructing this disease [1]. Many Latin American countries are located in tropical or subtropical regions, which coincide with the habitat of the *Aedes* mosquito. Therefore, these countries face the common problem of spreading the diseases transmitted by *Aedes* mosquitoes, especially dengue, which accounts for approximately 90% of all cases on the American continent [2].

A countermeasure to prevent further expansion of this disease is massive fumigation to exterminate these species of mosquitoes. However, massive fumigation causes irreversible ecological damage in addition to economic burden. Therefore, in many countries, fumigation of a specific area is only permitted when human dengue cases are reported. This is because detecting infected mosquitoes is only possible through molecular biology in specialized laboratories, which is time-consuming and expensive [3].

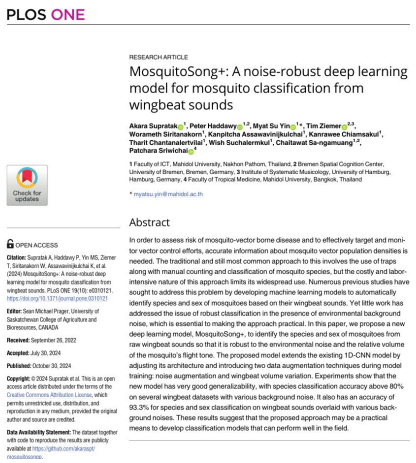
Considering the above-mentioned situations, the final objective of this paper is to propose a technological solution to detect dengue-infected *Aedes aegypti* mosquitoes in the field. The detection must be rapid to avoid a possible virus spread between humans, and detection accuracy must be as high as possible, with the lowest uncertainty, to perform effective fumigation. Effective fumigation minimizes the impact on ecosystems and reduces the economic burden. This research could potentially contribute to two of the United Nations Sustainable Development Goals (SDGs): Goal 3, "Good health and well-being," and Goal 15, "Life on Land."

Some scientists have reported that the behavior of infected mosquitoes has changed compared to that of healthy mosquitoes [4-6]. For instance, the authors of [4] and [5] reported that the infected female *Aedes* mosquitoes bite more frequently to obtain the same quantity of blood required by her eggs. The authors of [6] also observed that infected mosquitoes exhibited increased locomotor activity compared to healthy mosquitoes. These observations related to the change in behavior of infected mosquitoes suggest a possible hypothesis about the change in flight form of infected mosquitoes, and the possibility that infected mosquitoes can be detected by analyzing their wingbeat signals, which directly reflect the flight form of mosquitoes.

All living organisms, even those of the same species, have individual differences. The differences between individual mosquitoes lie in their birth season, feeding habits during their immature stages, and environmental factors such as temperature and humidity. These individual differences are

Memória temporal

Supratak · 2024



PLOS ONE

RESEARCH ARTICLE

MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds

Akara Supratak[1], Peter Haddawy[1,2], Myat Su Yin[1]*, Tim Ziemer[2,3], Worameth Siritanakorn[1], Kanpitcha Assawavinijkulchai[1], Kanrawee Chiamsakul[1], Thanit Chantanalertvilai[1], Wish Suchalermkul[1], Chaitawat Sa-ngamuang[1,2], Patchara Sriwichai[4]

1 Faculty of ICT, Mahidol University, Nakhon Pathom, Thailand, **2** Bremen Spatial Cognition Center, University of Bremen, Bremen, Germany, **3** Institute of Systematic Musicology, University of Hamburg, Hamburg, Germany, **4** Faculty of Tropical Medicine, Mahidol University, Bangkok, Thailand

* myatsu.yin@mahidol.ac.th

Abstract

In order to assess risk of mosquito-vector borne disease and to effectively target and monitor vector control efforts, accurate information about mosquito vector population densities is needed. The traditional and still most common approach to this involves the use of traps along with manual counting and classification of mosquito species, but the costly and labor-intensive nature of this approach limits its widespread use. Numerous previous studies have sought to address this problem by developing machine learning models to automatically identify species and sex of mosquitoes based on their wingbeat sounds. Yet little work has addressed the issue of robust classification in the presence of environmental background noise, which is essential to making the approach practical. In this paper, we propose a new deep learning model, MosquitoSong+, to identify the species and sex of mosquitoes from raw wingbeat sounds so that it is robust to the environmental noise and the relative volume of the mosquito's flight tone. The proposed model extends the existing 1D-CNN model by adjusting its architecture and introducing two data augmentation techniques during model training: noise augmentation and wingbeat volume variation. Experiments show that the new model has very good generalizability, with species classification accuracy above 80% on several wingbeat datasets with various background noise. It also has an accuracy of 93.3% for species and sex classification on wingbeat sounds overlaid with various background noises. These results suggest that the proposed approach may be a practical means to develop classification models that can perform well in the field.

Check for updates

OPEN ACCESS

Citation: Supratak A, Haddawy P, Yin MS, Ziemer T, Siritanakorn W, Assawavinijkulchai K, et al. (2024) MosquitoSong+: A noise-robust deep learning model for mosquito classification from wingbeat sounds. PLoS ONE 19(10): e0310121. https://doi.org/10.1371/journal.pone.0310121

Editor: Sean Michael Prager, University of Saskatchewan College of Agriculture and Bioresources, CANADA

Received: September 26, 2022

Accepted: July 30, 2024

Published: October 30, 2024

Copyright: © 2024 Supratak et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: The dataset together with code to reproduce the results are publicly available at https://github.com/akarasupt/mosquitosongs.

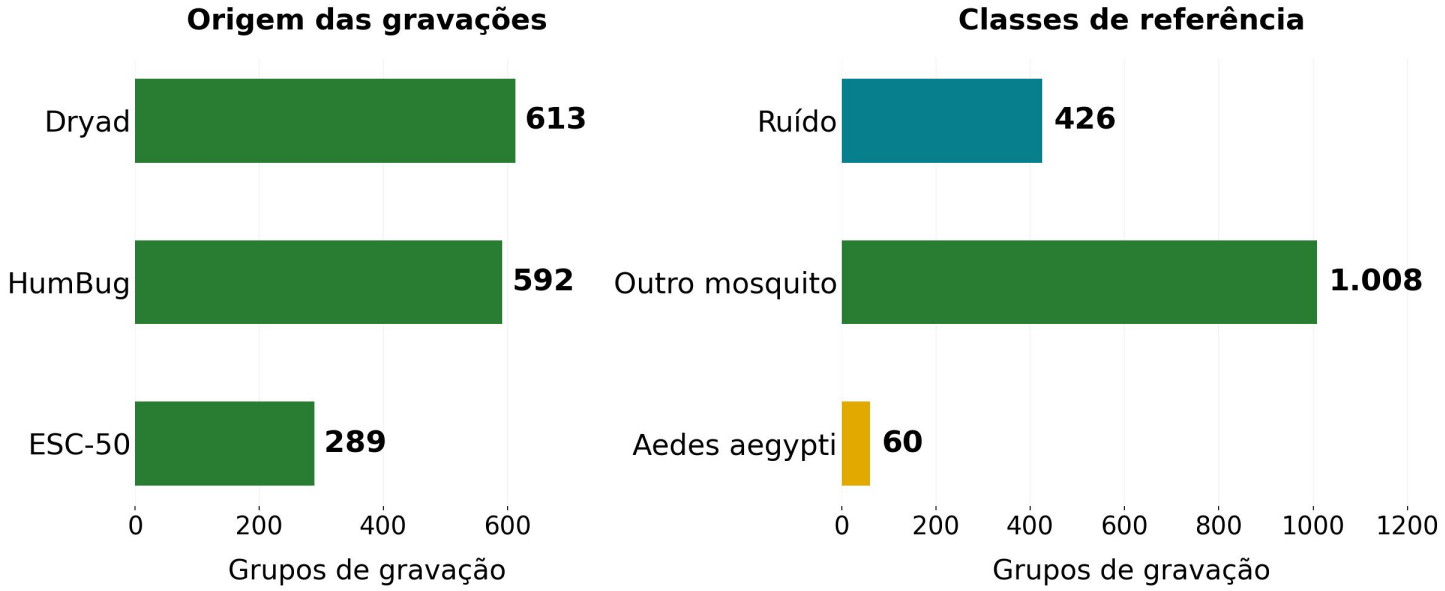
Robustez ao ruído

Técnicas adaptadas à tarefa ternária; resultados medidos neste projeto.

INSTITUTO FEDERAL
Santa Catarina

Separar grupos permite avaliar o aprendizado

Dryad, HumBug e ESC-50 · 42.029 janelas de 0,992 s

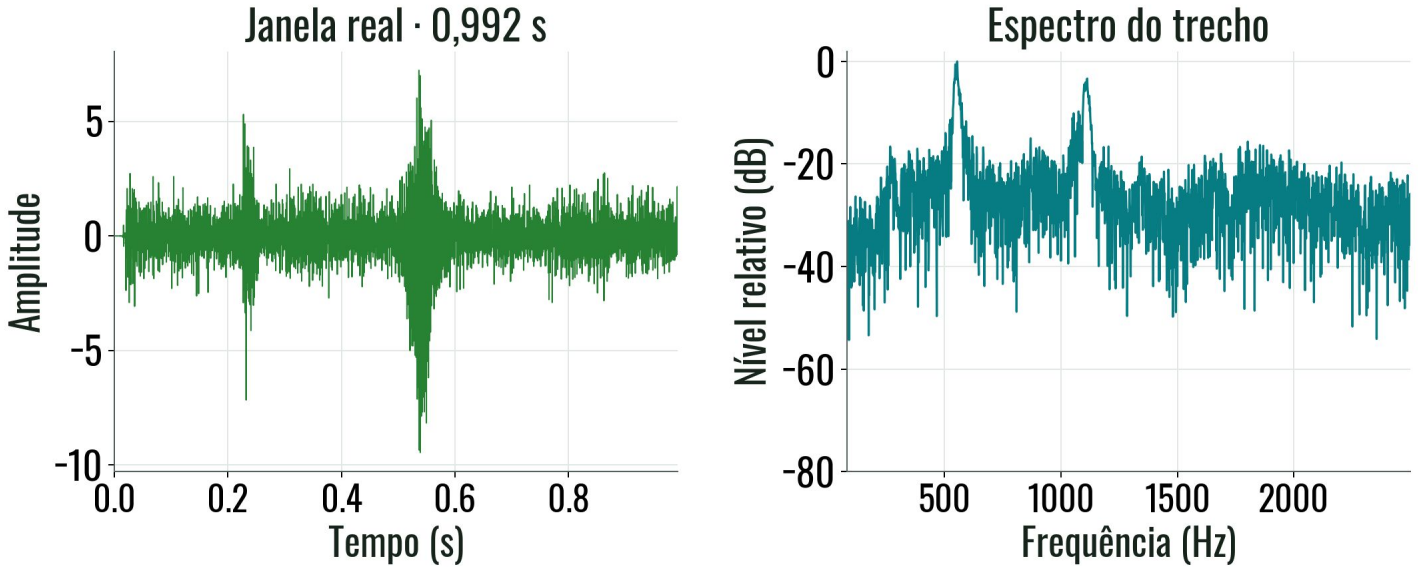


5 dobras por grupo: 4 ajustam; 1 testa · cada grupo testado uma vez.
Dryad/HumBug: mosquitos e fundos · ESC-50: sons ambientais.

INSTITUTO FEDERAL
Santa Catarina

A FFT revela as frequências presentes no trecho

Um mesmo trecho real de voo: amplitude no tempo e espectro.

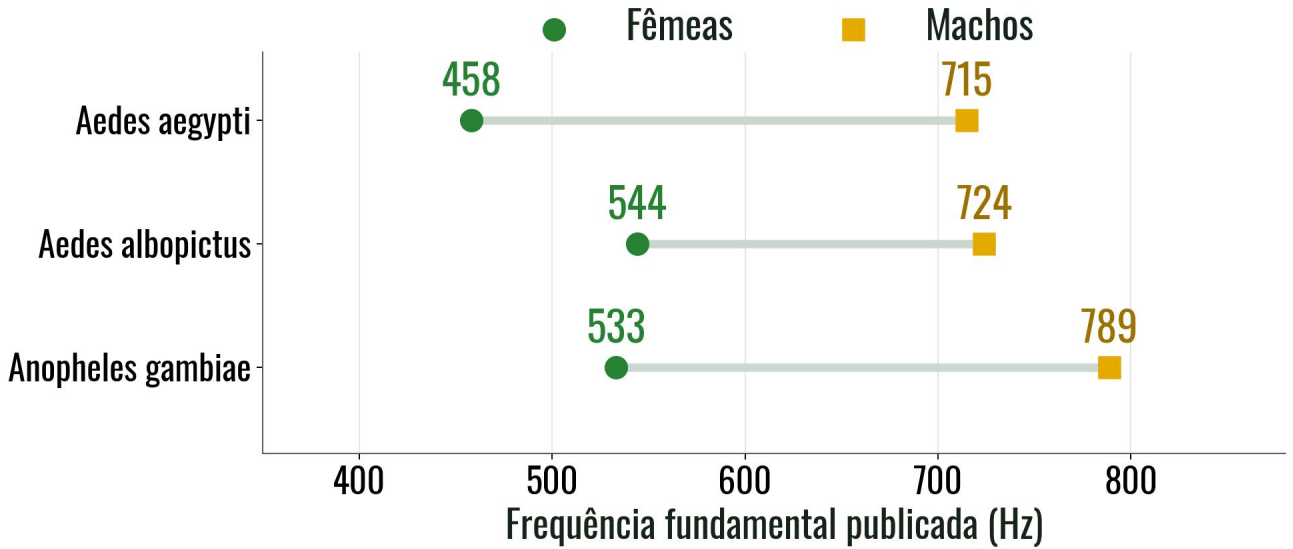


FFT decompõe o sinal temporal em componentes de frequência.

INSTITUTO FEDERAL
Santa Catarina

O voo contém uma fundamental e seus harmônicos

Médias publicadas a 25 °C para fêmeas e machos · Sinka et al. (2021)

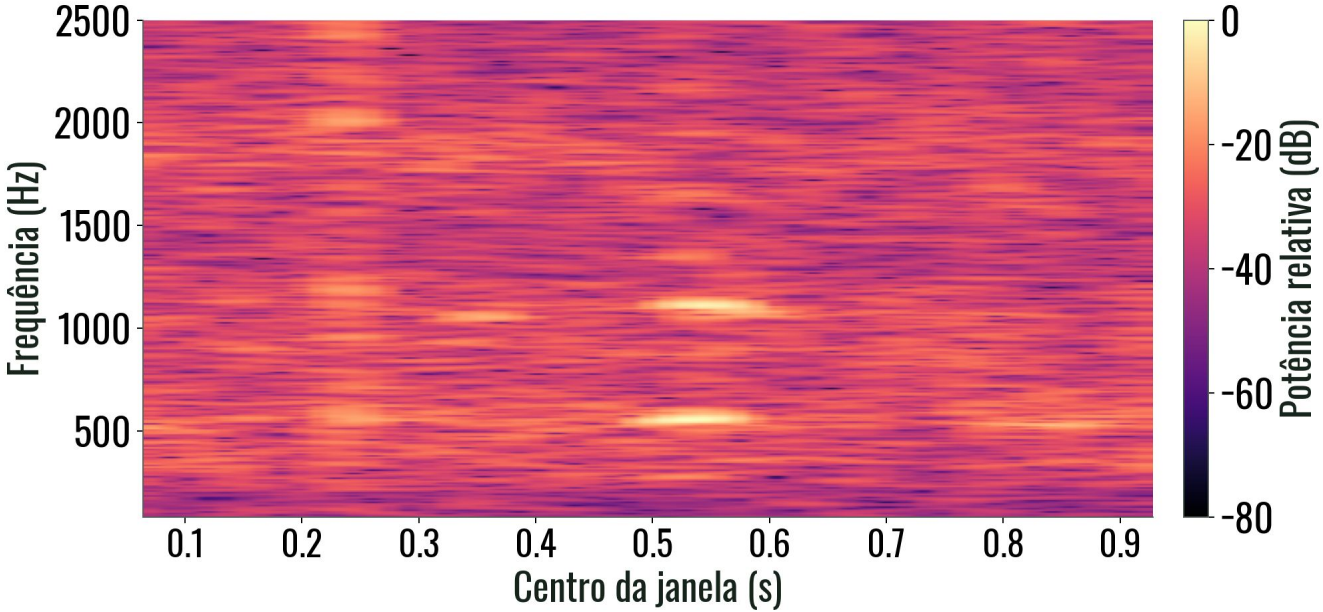


Fundamental: ritmo básico · harmônicos: múltiplos dessa frequência.

INSTITUTO FEDERAL
Santa Catarina

A STFT localiza as frequências ao longo do voo

A STFT repete a FFT em janelas para preservar a posição no tempo.

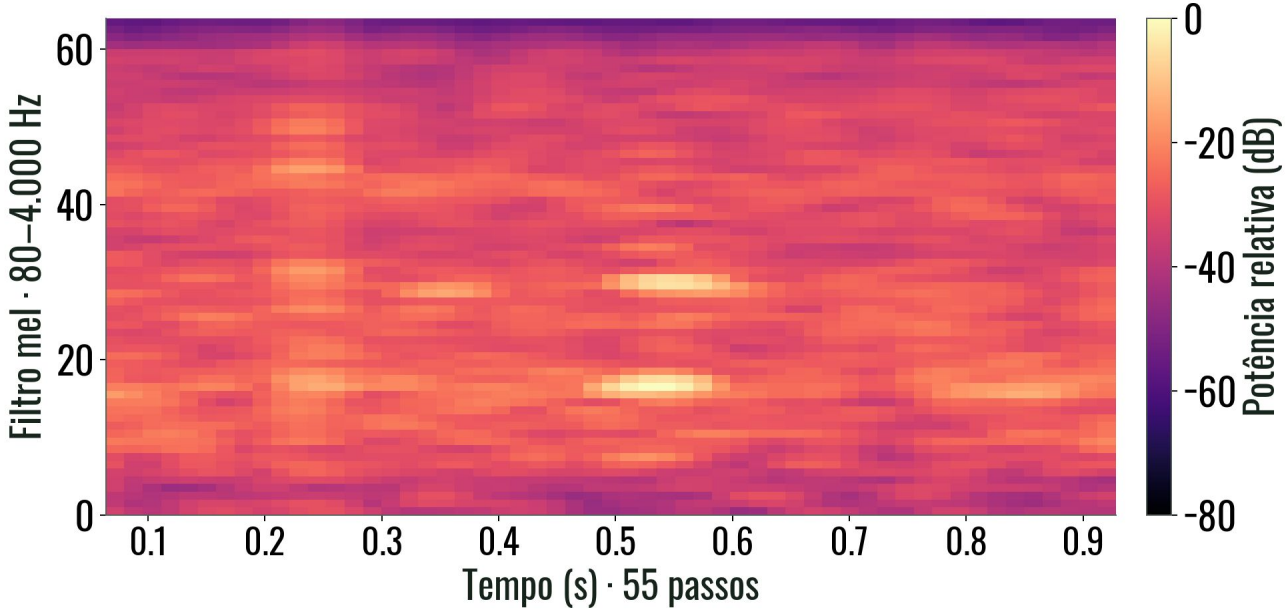


Hann 128 ms · avanço 16 ms · FFT 1.024 · áudio a 8 kHz.

INSTITUTO FEDERAL
Santa Catarina

As bandas mel formam a sequência de entrada

Mel resume a energia em bandas: uma sequência compacta para a rede.

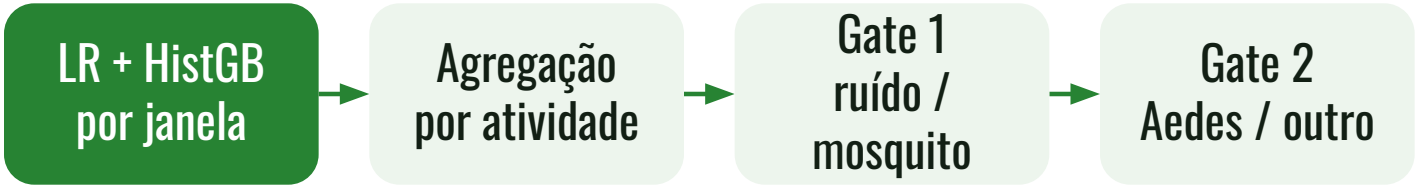


64 bandas de 80–4.000 Hz · 55 quadros por trecho · energia em dB.

INSTITUTO FEDERAL
Santa Catarina

A cascata organiza a decisão em duas etapas

A referência combina atributos acústicos antes de aplicar dois limiares.



Dois modelos, uma decisão por gravação

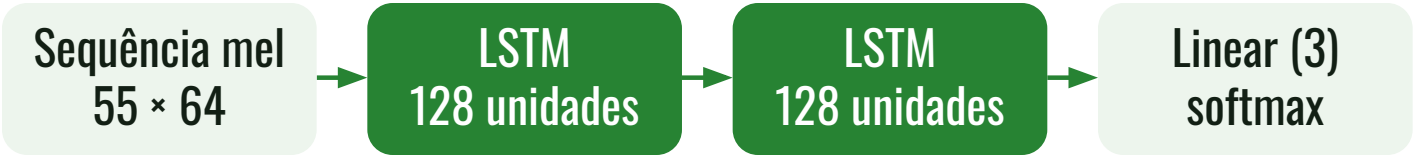
LR: regressão logística · HistGB: árvores em boosting · gate: limiar.

INSTITUTO FEDERAL
Santa Catarina

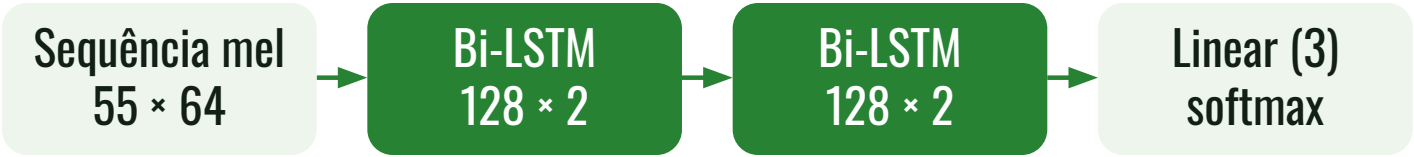
LSTM e Bi-LSTM aprendem relações no tempo

A memória temporal relaciona quadros mel de um mesmo trecho.

LSTM · memória do início até o fim



Bi-LSTM · contexto nos dois sentidos

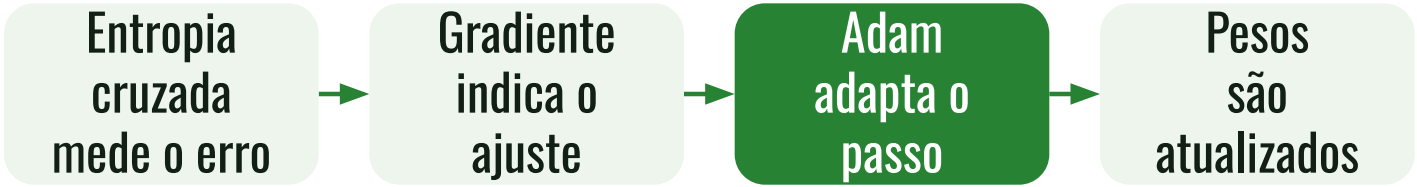


Duas camadas · 128 unidades por direção · dropout 0,20 · três saídas.

INSTITUTO FEDERAL
Santa Catarina

Adam transforma o erro em ajustes nos pesos

O erro gera gradientes; o otimizador atualiza os parâmetros.



Taxa 0,0006 · lote de 256 janelas
Clipping = 3 · até 30 épocas

Clipping limita gradientes altos e ajuda a estabilizar o treino.

INSTITUTO FEDERAL
Santa Catarina

Regularização e validação controlam o ajuste

1 **Dropout reduz dependências entre unidades**

Zera 20% das ativações entre camadas.

2 **Amostragem equilibra as três classes**

Classes equilibradas; peso por gravação com $\gamma = 0{,}50$.

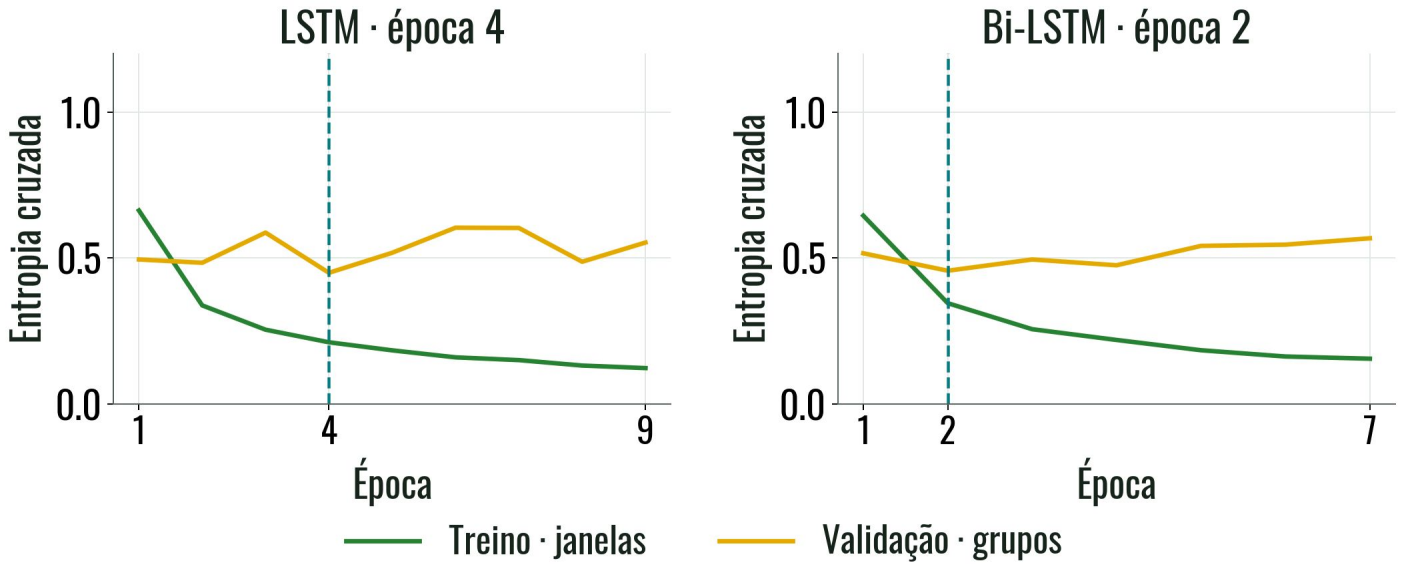
3 **Validação escolhe quando parar**

Guarda a menor perda; paciência de 5 épocas.

MC dropout: 100 passagens medem discordância na inferência.

A validação seleciona o checkpoint de cada rede

Época: um ciclo de amostragem e atualização dos pesos.

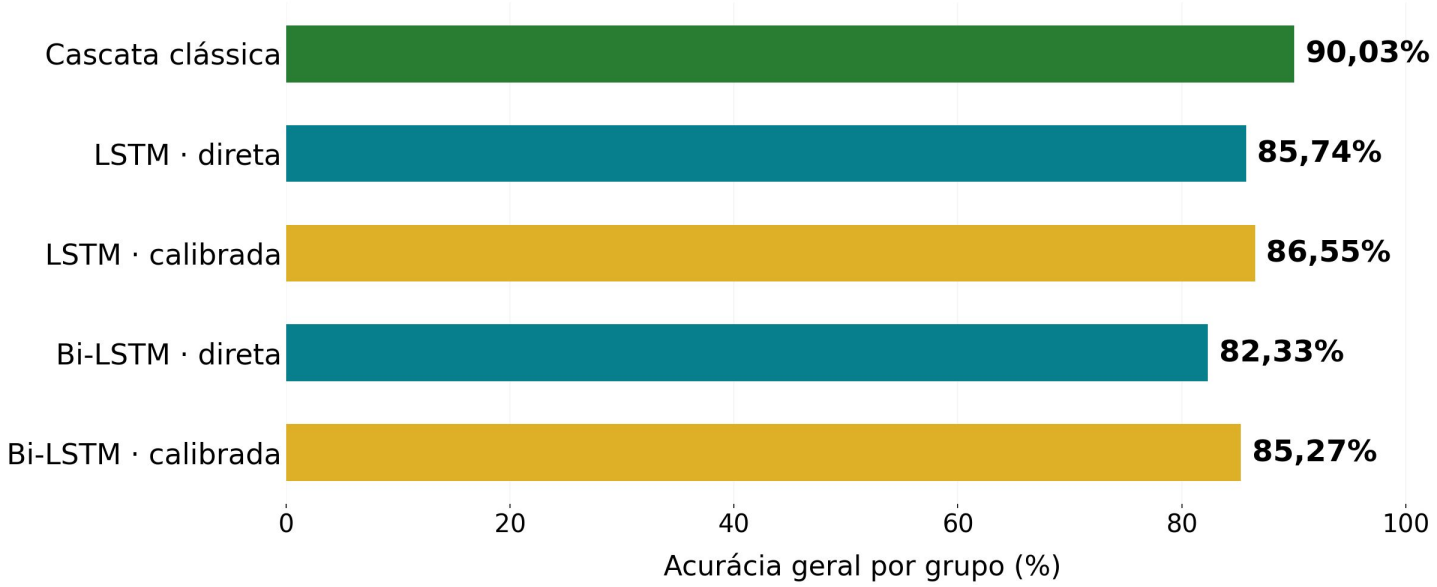


LSTM: melhor época 4 de 9 observadas · Bi-LSTM: 2 de 7 · contexto final.

INSTITUTO FEDERAL
Santa Catarina

A cascata teve o maior acerto geral

Acurácia = classificações corretas ÷ todos os grupos avaliados.



Mesmos 1.494 grupos · direta: maior probabilidade · calibrada: dois limiares.

INSTITUTO FEDERAL
Santa Catarina

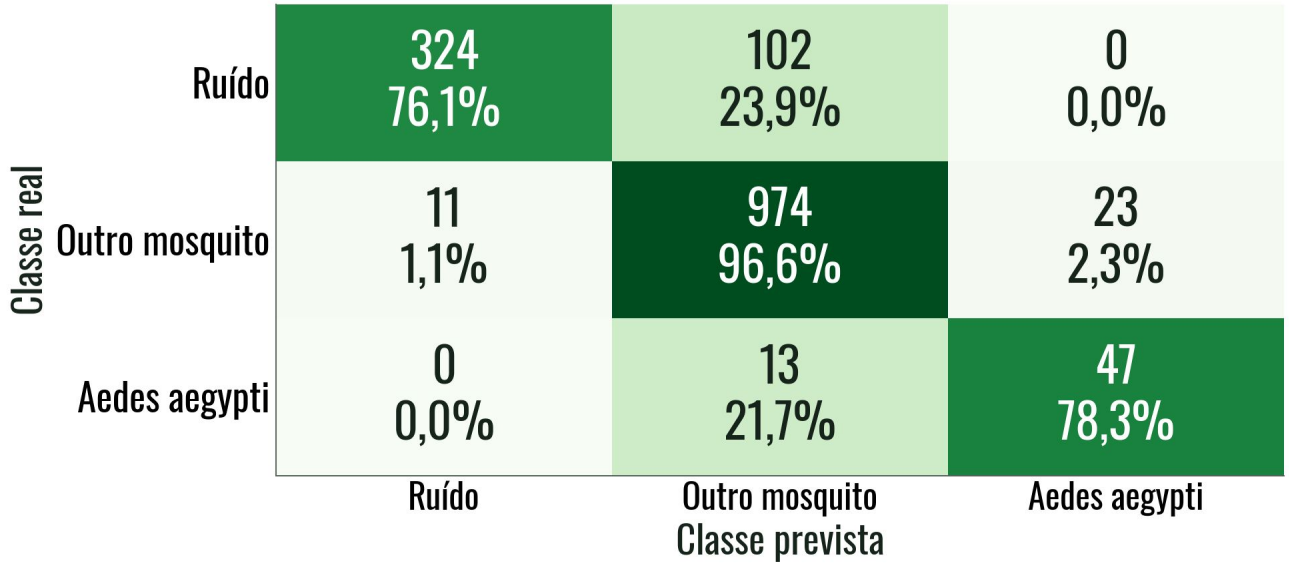
Detectar mais Aedes muda o número de alarmes

Modelo / regra	Sensibilidade Aedes	FPR ruído para Aedes	FPR outro para Aedes
Cascata clássica	78,33%	0,00%	2,28%
LSTM · direta	86,67%	3,76%	3,97%
LSTM · calibrada	33,33%	0,47%	0,60%
Bi-LSTM · direta	88,33%	4,93%	6,25%
Bi-LSTM · calibrada	28,33%	0,00%	0,40%

Sensibilidade: TP/60 · FPR ruído: FP/426 · FPR outro: FP/1.008.

A matriz mostra quais classes foram confundidas

Linhas: classe real · colunas: previsão · diagonal: acertos.



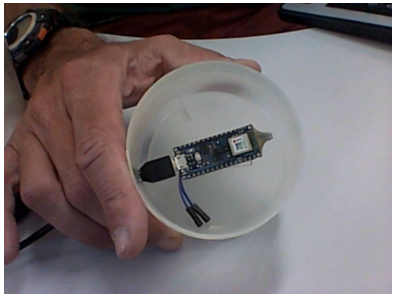
Aedes: 47/60 detectados · precisão: 47/70 alertas = 67,14%.

INSTITUTO FEDERAL
Santa Catarina

O protótipo prepara a coleta de uma base própria

Microfone capta o voo.
USB leva o áudio ao PC.
WAV preserva o sinal.
Identificar espécie e sessão.
Incluir outros e ruído.
Separar indivíduos no teste.





Montagem e placa · fotos fornecidas pelo autor

Próximo passo: coletar, rotular e validar.

INSTITUTO FEDERAL
Santa Catarina

Referências e material de apoio

Mukundarajan et al. (2017) · captura com celulares

Torres et al. (2025) · memória temporal e MC dropout

Supratak et al. (2024) · CNN e robustez ao ruído

Sinka et al. (2021) · frequências e HumBug

Código e resultados · github.com/Lciarallo/mosquito-wingbeat

Marca: IFSC · fotos: CDC, SES-RS, SESA-ES e acervo do projeto

Conceitos, contas e fontes completas no guia e nas notas.

INSTITUTO FEDERAL
Santa Catarina

O estudo compara desempenho e orienta a coleta

1

O que foi demonstrado

Cascata: 90,03%; redes diretas detectam mais Aedes.

2

Onde a conclusão se aplica

Corpus conhecido; sessões e dispositivos podem se repetir.

3

Próxima validação

Coleta própria, sessões novas e detector congelado.

MosquitoSong externo: 0/4 grupos Aedes · redes sem porte embarcado validado.

INSTITUTO FEDERAL
Santa Catarina